

FSOC VIRTUAL CAMERA TRACKING SYSTEM

Coarse Alignment - Performance Report

benchmark video: synthetic_fsoc_beacon.mp4

UTC 2026-09-24T20:00:03

1. RUN CONFIGURATION

Frames Processed

450

Duration

1.89 s

Processing FPS

237.6

Avg Processing

1.73 ms

Max Processing

3.24 ms

2. TRACKING PERFORMANCE

Acquisition Time

0.03 s

<= 2 s

Average Tracking Error

2.68 px

Maximum Tracking Error

6.08 px

<= 10 px

RMSE Tracking Error

2.97 px

Centroid Error Avg/Max

2.68 / 6.08 px

ground truth only

Lock Retention

100.00 %

Target Loss

0.00 %

< 5 %

Re-acquisition Avg/Max

—

<= 1 s

Successful Re-acq.

0

0 loss events

3. PS REQUIREMENT CHECK

Parameter	Measured	Requirement	Status
Acquisition Time	0.03	<= 2.0 s	PASS
Tracking Error (Max)	6.08	<= 10 px	PASS
Center Error (Final)	9.22	<= 10 px	PASS
Target Loss	0.00 %	< 5%	PASS
Re-acquisition Time	0.13	<= 1.0 s	PASS
Processing Speed	237.6 FPS	>= 20 FPS	PASS

Ground truth was available: centroid accuracy is measured.

OVERALL: PASS

Source: benchmark video: synthetic_fsoc_beacon.mp4 | Frames: 450 | FPS: 237.6